Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/15/2021
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	Bacterial, Archaeal, & Fungal Diversity of the ISS		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Organization Name:	NASA Jet Propulsion Laboratory		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	03/31/2021
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Loftus, David	**Contact Phone:**	650-604-1011
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Flight Program:	ISS		
Flight Assignment:	NOTE: Title change in May 2015 to "Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System" from "Bacterial, Archaeal, & Fungal Diversity of Kibo-JEM Surfaces," per A. Chu/ NASA ARC via NASA JPL Rev. A to Task Order (Ed., 9/24/15) NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 12/16/21) NOTE: End date changed to 10/01/2021 per F. Hernandez/ARC (Ed., 4/19/21) NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 1/18/21) NOTE: End date changed to 12/31/2020 per F. Hernandez/ARC (Ed., 12/6/19) NOTE: Extended to 12/31/2019 and project number changed to 47-700062 under Task Order NNN17D016T per F. Hernandez/ARC (Ed., 12/17/18) NOTE: Extended to 9/30/2018 per F. Hernandez/ARC (Ed., 3/26/18) NOTE: Extended to 3/31/2018 per F. Hernandez/ARC (Ed., 8/31/17) NOTE: Extended to 6/30/2017 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:	September 2017 report: Parag Vaishampayan in no longer Co-Investigator on this project.		

COI Name (Institution):	
Grant/Contract No.:	47-700062 under Task Order NNN17D016T ; 19-12829-27 under Task Order NNN13D111T
Performance Goal No.:	
Performance Goal Text:	
Task Description:	May 2015 Revision to Task Agreement (Ed. note: information received September 2015). Due to problems obtaining samples from Kibo-JEM (Japanese Experiment Module), the project has been modified with a new title, "Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System." The revised scope will allow the Jet Propulsion Laboratory (JPL) to undertake a 28-month flight study to examine the microorganisms associated with HEPA filters. The ISS-HEPA project will develop a comprehensive census of the microbial population on the HEPA filter collected from various locations of the International Space Station (ISS), using genetic biosignatures and computational modeling. This microbial census will provide NASA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL team will determine which microbial taxa pose particular threats to crew health. Original Task Description: Space microbiology research working groups in Japan, authorized by the Japan Aerospace Exploration Agency (JAXA), submitted proposals in response to the JAXA Announcement of Opportunity, 2012. The JAXA proposal that is managed by Principal Investigator Prof. Masao Nasu, Osaka University, contained several subgroups, including subgroup 4, headed by Dr. Takashi Yamazaki, JAXA. This subgroup 4 will conduct an investigation on "Analysis of microbial dynamics in the space habitation system by using new molecular microbial ecological techniques and also monitoring microbial contamination from air and surfaces of the 'Kibo' Japanese Experiment Module (Kibo-JEM)." The JAXA proposal submitted by subgroup 4 identifies the Jet Propulsion Laboratory (JPL) team headed by Dr. Kasthuri Venkateswaran as a collaborator to jointly explore the investigation. The Bacterial, Archaeal, and Fungal Diversity of Kibo-JEM Surfaces project (JPL-JAXA project) will undertake a three-year flight study to examine the changes in the microorganism population on the Kibo-JEM surfaces. The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health. Specific JPL-JAXA project goals will enable us to resolve applicable Human Research Program (HRP) integrated research plan (IRP) risks: (1) determine if changes in Kibo-JEM microbiome are occurring during a spaceflight mission and if so, (2) determine whether any changes noted would create a greater health risk for the crew. There are three aims proposed in this JPL-JAXA project and study of the Kibo-JEM microbiome. Through optimizations and comparative evaluations of methods designed to collect, process, analyze, and archive biomolecules sampled, the JPL-JAXA team will increase downstream sensitivity of laboratory assays as well as enhance the current understanding of the population dynamics of the Kibo-JEM module (Aim 1). The JPL-JAXA team will adapt molecular technologies currently in use with the National Institute of Health (NIH) – Human Microbiome Program (HMP) Roadmap—such as rRNA gene-variable region tag sequencing, whole genome shotgun sequencing (WGS), and DNA microarrays, as well as cloning and sequencing of 16S ribosomal DNA (rDNA)—to comprehensively elucidate the microbial diversity profiles associated with the Kibo-JEM surfaces (Aim 2). The final project deliverable to NASA will be the Kibo-JEM database, a compilation of all of the genomic sequences and genetic information of microbes encountered on and within the Kibo-JEM habitat (Aim 3). Computational modeling of the Kibo-JEM microbial population dynamics inferred from the database will then be carried out to predict the complete spectrum of organisms present. The JPL-JAXA project will compare microbial diversity, microbial characteristics, and functional capabilities of the microorganisms associated with the Kibo-JEM. The resulting database will enable HRP to evaluate the effect of spaceflight radiation and microgravity on the Kibo-JEM microbiome. The JPL-JAXA project in combination with results from studies of crew susceptibility will be used to determine the risk of microbiologically induced adverse health effects during a spaceflight mission. This assessment will allow HRP to determine if current operational and engineering controls used to mitigate microbiological risks are adequate or whether additional countermeasures should be developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This microbial census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health.
Task Progress:	During this task, an inventory of microorganisms isolated from the HEPA filters and environmental surfaces of the International Space Station (ISS) as well as crew resupply vehicles (CRV) during multiple NASA projects have been archived, and inventoried under Space Biology Microbial Culture Collection (SBMCC). The whole-genome sequences (WGS) of SBMCC were generated and made available to the public. In addition to multiple copies of same microbial species, the SBMCC consists of several newly identified microorganisms. The WGS of the characterized bacterial isolates include Lactobacillales, Enterobacteriales, Bacillaceae, Agrobacterium, Klebsiella, Methylobacterium, and Staphylococcus. Moreover, isolates of fungi have also been sequenced, belonging to the members of fungal phyla Ascomycota and Basidiomycota. This SBMCC archival of the ISS and CRV will aid in the ongoing characterization of short sequence, shot-gun metagenome surveys of the microbial diversity present in these built environments. The availability of these new WGS will also enable genetic comparisons between microorganisms sampled from Earth-based environments with those sampled from spacecraft surfaces of the ISS and CRV. Such comparisons will help to elucidate genetic adaptations that allow some of these microorganisms to become more virulent and allow all of them to survive the harsh environment of space (e.g., ionizing radiation, microgravity, desiccation, and oxidative stress). Additionally, when annotated, functional

	analyses of these WGS can help to identify targeted strategies for the control of problematic microorganisms that may degrade metals and plastics, become potentially pathogenic and harm astronauts, or obstruct water supply due to their biofilm forming capabilities in the engineering systems of the closed habitat.
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